

Define fidelity criteria of image compression. Draw a model of image compression.  
Briefly describe JPEG, JPEG 2000, PDF and BMP standards for image compression.

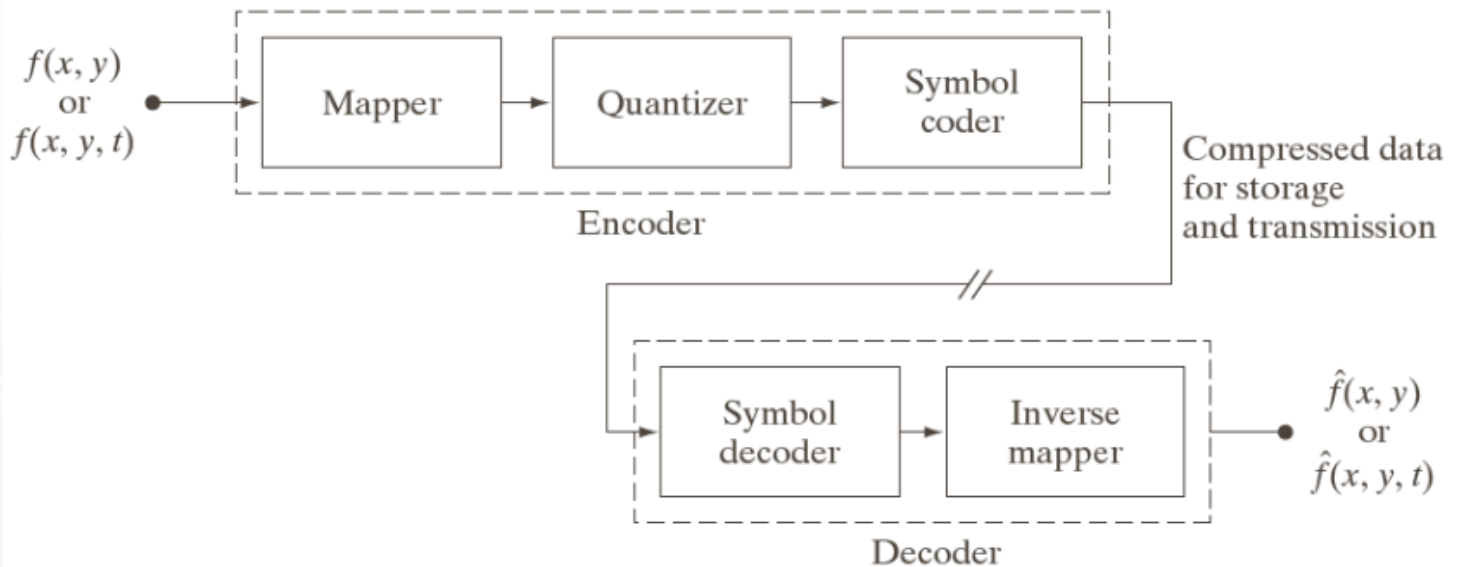
Fidelity criteria measure how well a reconstructed image matches the original.

Common Fidelity Criteria:

1. Mean Square Error: Average squared difference between original and compressed pixels, lower values signify higher quality.
2. Peak Signal-to-Noise Ratio: Ratio of maximum signal power to compression noise, higher values indicate better quality.
3. Subjective Visual Quality: Human assessment of whether image distortions are perceptible or acceptable.

Model of image compression:

# Image Compression Model



Briefly describe JPEG, JPEG 2000, PDF and BMP standards for image compression.

**JPEG** – Lossy compression using DCT on 8×8 pixel blocks. Great for photos, but causes blocking artifacts at high compression.

**JPEG 2000** – Improved successor using wavelet transforms. Supports lossy and lossless modes, no blocking artifacts, and transparency — but rarely adopted outside cinema and medical imaging.

**PDF** – A document format that embeds images using multiple codecs, optimizing compression per content type (photos, text, graphics).

**BMP** – A simple Microsoft format storing raw, largely uncompressed pixel data. Very large files, but trivial to parse and perfectly lossless.